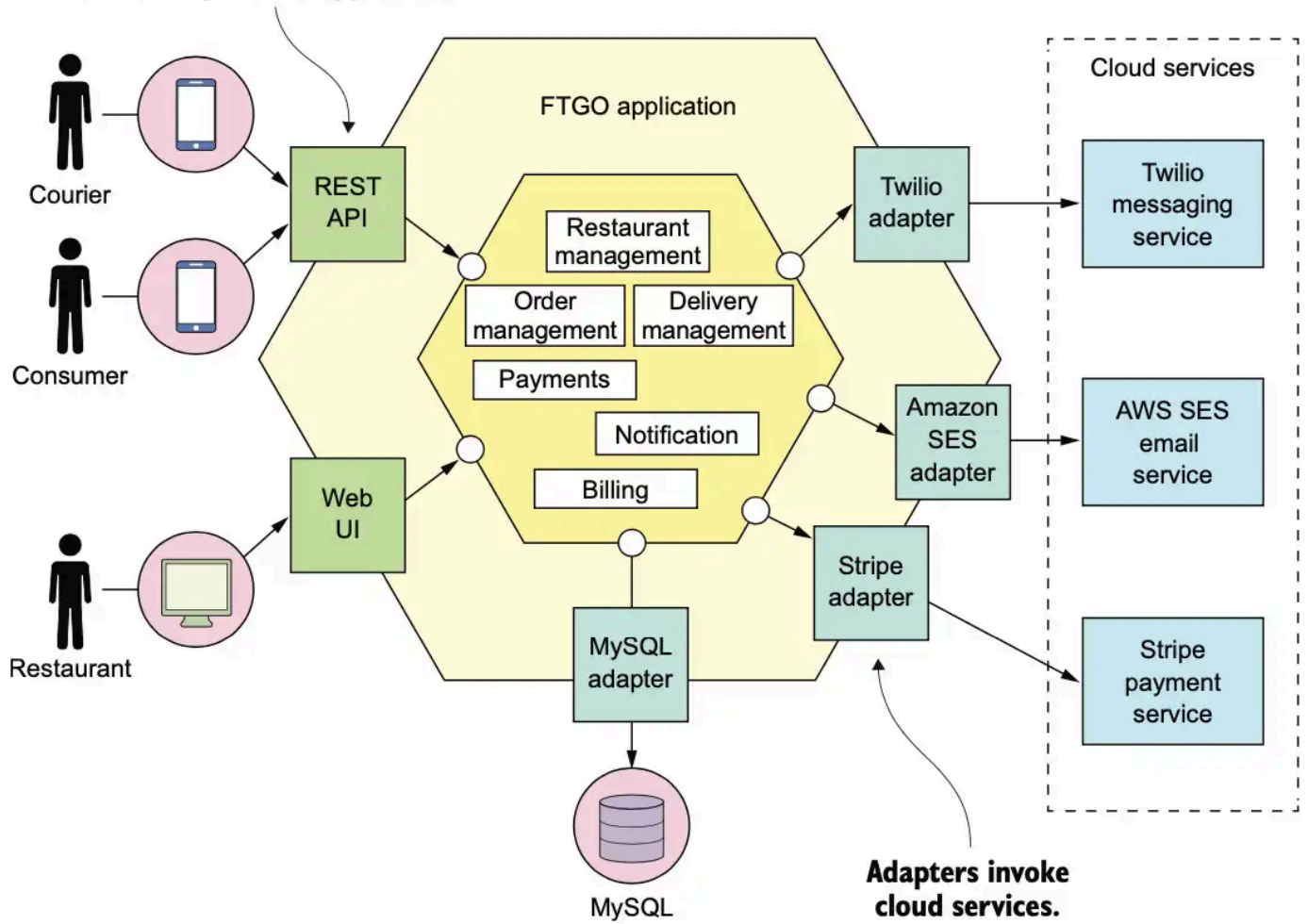


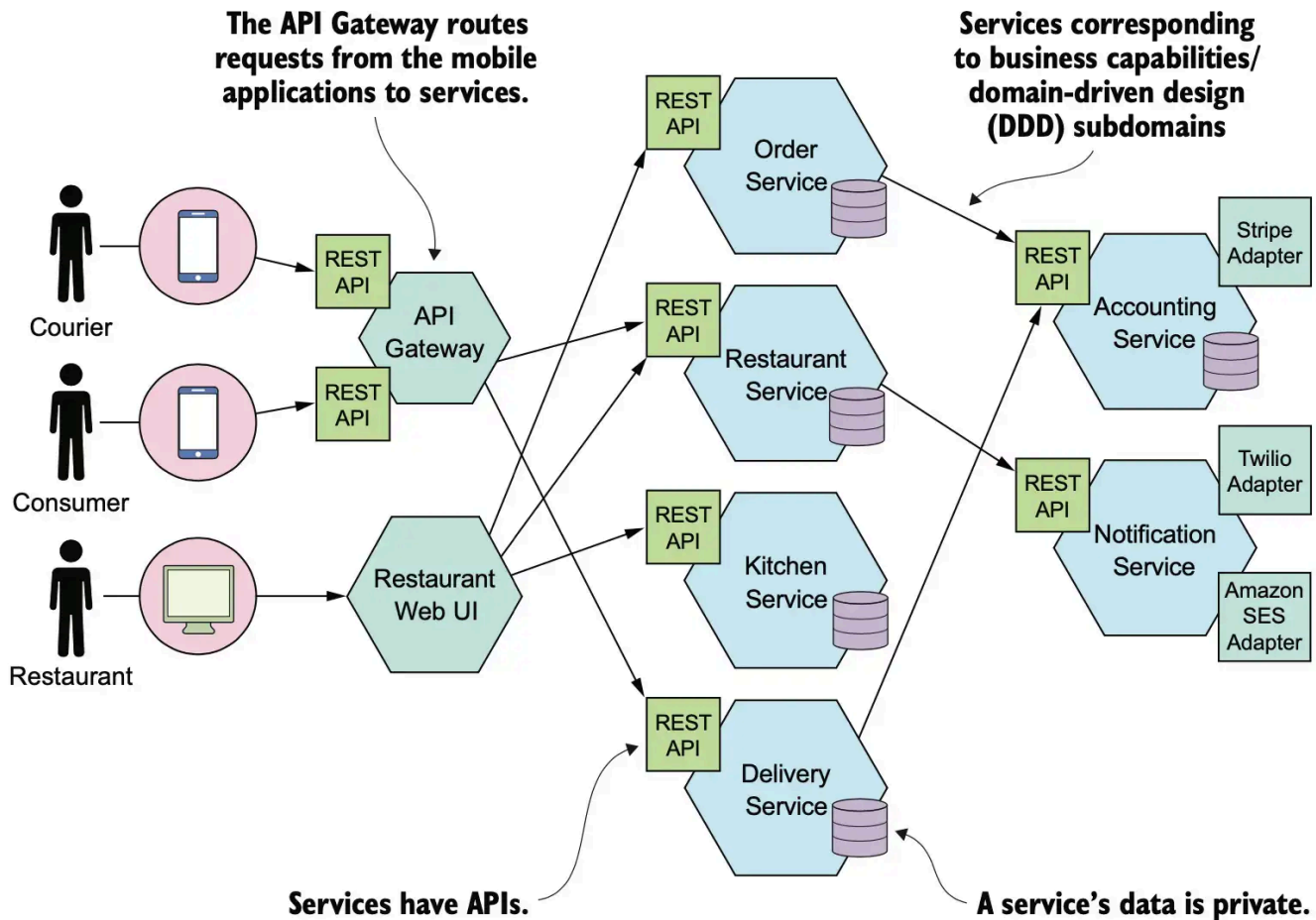
The Twelve-Factor App

The Twelve-Factor App is a methodology for building **modern, scalable, cloud-native applications**.

- Portability across environments
- Continuous deployment
- Scalability and resilience
- Maintainability in distributed systems

Invoked by mobile applications





I. Codebase

- One codebase tracked in version control
- Many deploys from the same codebase

Benefits:

- Traceability and collaboration
- Simplified CI/CD
- Consistent releases across environments

II. Dependencies

- Explicitly declare and isolate dependencies

Key Ideas:

- Use dependency managers (e.g., Maven, Gradle, pip)
- Avoid relying on system-level packages
- Ensures reproducibility and portability

III. Config

- Store configuration in the environment

Examples:

- Database URLs
- API keys
- Credentials

Benefits:

- Separation of code and configuration
 - Easier deployment across environments
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IV. Backing Services

- Treat external services as attached resources (*An attached resource is something the application does not own or embed internally, but connects to at runtime through configuration.*)

Examples:

- Databases
- Message brokers
- Caches

Key Principle:

- Easily replace services without code changes
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V. Build, Release, Run

- Strict separation of stages

Stages:

1. Build → compile and package
2. Release → combine build with config
3. Run → execute the application

Benefits:

- Repeatable deployments
 - Reduced risk of errors
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VI. Processes

- Execute the app as one or more stateless processes

Key Concepts:

- No shared memory
- State stored in external services

Advantages:

- Scalability
 - Fault tolerance
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VII. Port Binding

- Export services via port binding

Implication:

- Application is self-contained
 - No external web server required
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VIII. Concurrency

- Scale out via process model

Example:

- Multiple instances behind a load balancer

Benefits:

- Elastic scalability
 - Improved resilience
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IX. Disposability

- Fast startup and graceful shutdown

Goals:

- Quick scaling
 - Resilience to failures
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X. Dev/Prod Parity

- Keep development, staging, and production similar

Focus Areas:

- Time
- Personnel
- Tools

Benefits:

- Reduced deployment issues
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XI. Logs

- Treat logs as event streams

Practices:

- Do not manage log files
 - Send logs to centralized systems
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XII. Admin Processes

- Admin tasks must:
- Use the same codebase
- Use the same configuration
- Run in the same environment

Examples:

- Database migrations
- Maintenance scripts

Benefits:

- Consistency with production environment
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Resources

- <https://www.12factor.net/>
- <https://www.youtube.com/watch?v=FryJt0Tbt9Q>